Ch1. Functions and Limits

§1.1 Functions and Their Representations

Definition A function is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E.

x : the independent variable
y : the dependent variable

$D = \mathrm{dom}(f)$: the domain of the function :
the set of all values that the independent variable is allowed to assume
the set of all possible values that make some values.

$\mathrm{ran}(f)$: the range of the function :
the set of all function values that the function assumes.
the set of all possible values
= $\{ f(x) : x \in \mathrm{dom}(f) \}$

Definition the graph of the function $y = f(x)$ with the domain D is the set of ordered pairs $\{ (x, f(x) : x \in D \}$

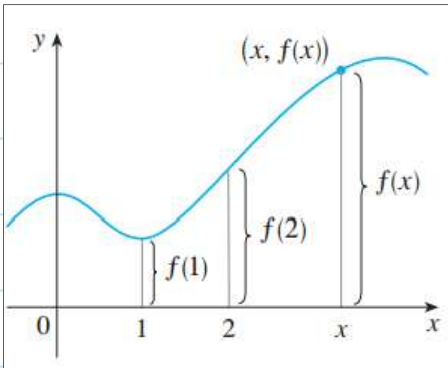


FIGURE 4

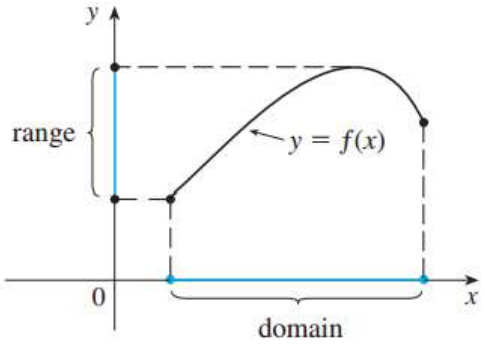


FIGURE 5

Example The graph of a function f is shown in Fig 6.
(a) Find the values of $f(1)$ and $f(5)$.
(b) What are the domain and range of f?

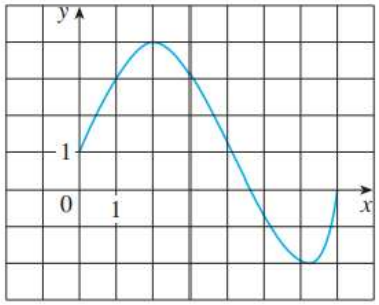


FIGURE 6

Example Find the domain of each function.

(a) $f(x)=\sqrt{x+2}$

(b) $g(x)=\dfrac{1}{x^2-x}$

Theorem [The Vertical Line Test]

A curve in the xy-plane is the graph of a function of x if and only if no vertical line intersects the curve more that once.

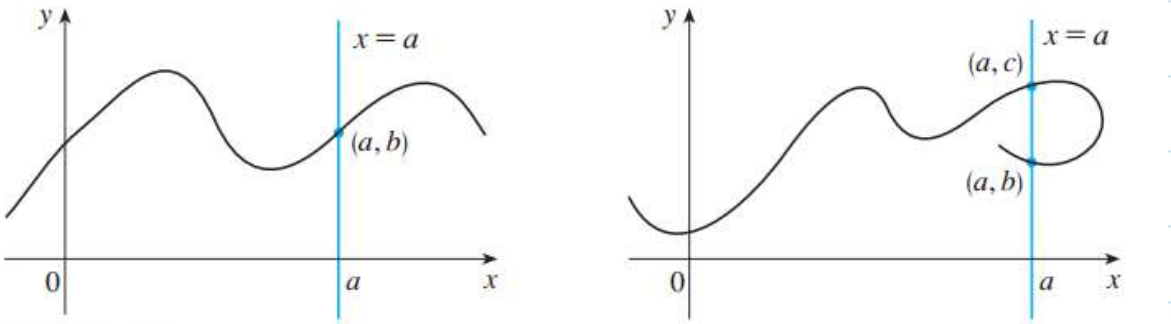


FIGURE 10

Example different [Piecewise Defined Function] : a function that is defined by different formulas in parts of its domain.

A function f is defined by

$$f(x)=\begin{cases}1-x & \text{if } x\le 1\\ x^2 & \text{if } x>1\end{cases}$$

Evaluate $f(0), f(1)$ and $f(2)$.

Example the absolute value of a number a, denoted by $|a|$

the absolute value function $f(x)=|x|$

$$|x|=\begin{cases}x & \text{if } x\geq 0\\ -x & \text{if } x<0\end{cases}$$

[Symmetry]

Defintion If a function $f(x)$ is called

1. an even function if it satisfies $f(x)=f(-x)$.;
2. an odd function if it satisfies $f(x)=-f(-x)$.

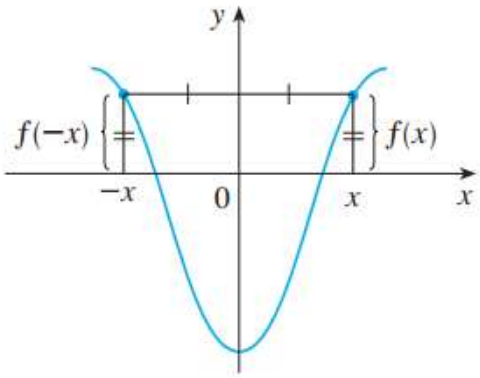


FIGURE 14 An even function

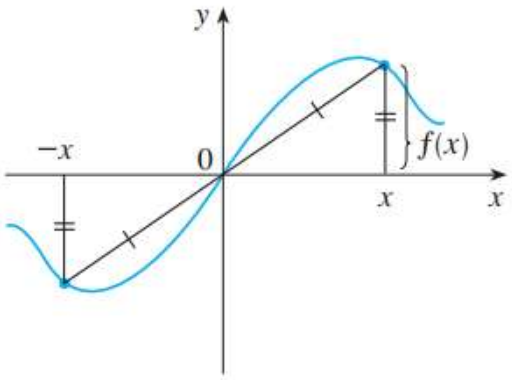


FIGURE 15 An odd function

Example Determine whether each of the following function is even, odd, or neither even nor odd.

(a) $f(x)=x^5+x$ (b) $g(x)=1-x^4$ (c) $h(x)=2x-x^2$

Definition A function is called increasing on an interval I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I

A function is called decreasing on an interval I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I

Example Let $f(x) = x^2$. On what interval is f increasing?

Ch1. Functions and Limits

§1.3 The Limit of a Function

[Intuitive Definition of a Limit]

Definition

Suppose $f(x)$ is defined when x is near the number a. (f is defined on some open interval that contains a, except possibly at a itself.) Then we write

$$\lim_{x \to a} f(x) = L$$

and say "the limit of $f(x)$, as x approaches a, equals L"

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a)) but not equal to a.

Notation

$\lim_{x \to a} f(x) = L$ === > $f(x) \to L$ as $x \to a$

" $f(x)$ approaches L as x approaches a. "

Example

Let $f(x) = x^2 - x + 2$.

f is defined for values of x near 2.

x	$f(x)$	x	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001

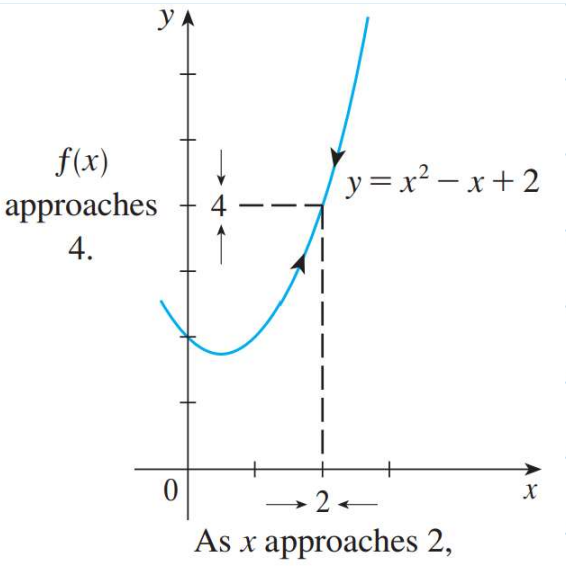


FIGURE 1

$$\lim_{x \to 2} x^2 - x + 2 =$$

Theorem If the functions f and g are equal for all x close to a, but not necessarily at $x = a$, then

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x)$$

whenever either limit exists.

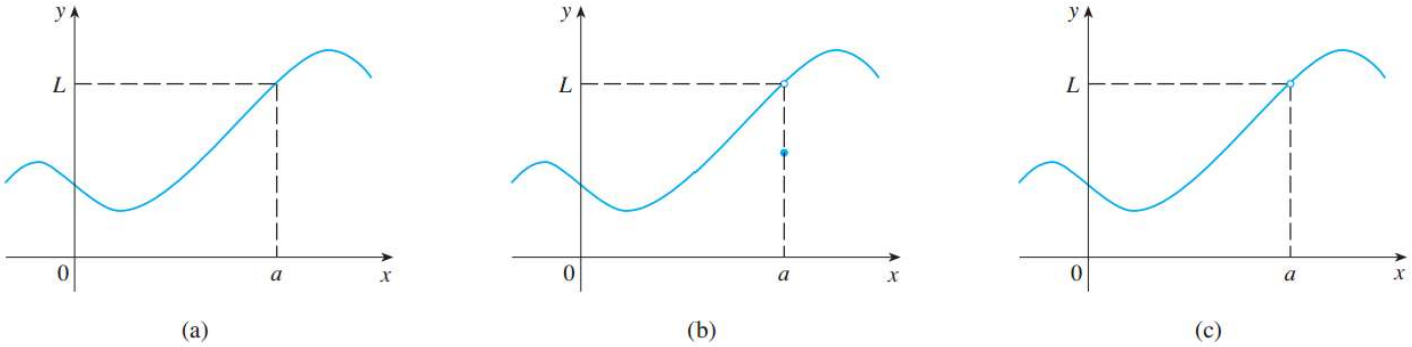


FIGURE 2
$\lim_{x \to a} f(x) = L$ in all three cases.

Example Guess the value of $\lim_{x \to 1} \dfrac{x-1}{x^2-1}$.

[One-Sided Limits]

Definition (a) [left limit or left-hand limit]

$\lim_{x \to a^-} f(x) = B$ means that we can make the values of $f(x)$ as close to B as we want by making sure that x is sufficiently closed to a and less than a.

(b) [right limit or right-hand limit]

$\lim_{x \to a^+} f(x) = A$ means that we can make the values of $f(x)$ as close to A as we want by making sure that x is sufficiently closed to a and greater than a.

$\lim_{x \to a} f(x) = A$ if and only if $\lim_{x \to a^-} f(x) = A = \lim_{x \to a^+} f(x)$

Example The Heaviside function H is defined by

$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$. Find the left limit and right limit of H as t approaches 0.

Example Find $\lim_{x \to 0} \frac{1}{x^2}$ if it exists.

x	$\frac{1}{x^2}$
± 1	1
± 0.5	4
± 0.2	25
± 0.1	100
± 0.05	400
± 0.01	10,000
± 0.001	1,000,000

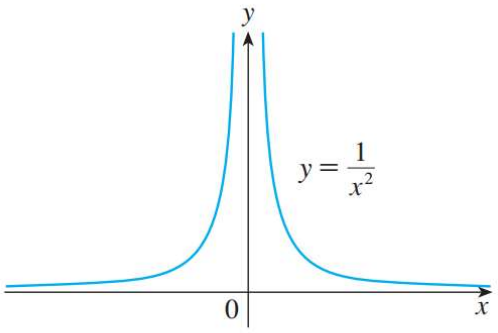


FIGURE 11

[Precise Definition of a Limit]

4 DEFINITION Let f be a function defined on some open interval that contains the number a, except possibly at a itself. Then we say that the **limit of $f(x)$ as x approaches a is L**, and we write

$$\lim_{x \to a} f(x) = L$$

if for every number $\varepsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$\text{if} \quad 0 < |x - a| < \delta \quad \text{then} \quad |f(x) - L| < \varepsilon$$

§1.4 Calculating Limits

LIMIT LAWS Suppose that c is a constant and the limits

$$\lim_{x \to a} f(x) \qquad \text{and} \qquad \lim_{x \to a} g(x)$$

exist. Then

1. $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$ — **sum law**
2. $\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$ — **difference law**
3. $\lim_{x \to a} [cf(x)] = c \lim_{x \to a} f(x)$ — **constant multiple law**
4. $\lim_{x \to a} [f(x)\, g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$ — **product law**
5. $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$ if $\lim_{x \to a} g(x) \neq 0$ — **quotient law**

6. $\lim_{x \to a} [f(x)]^n = \left[\lim_{x \to a} f(x)\right]^n$ where n is a positive integer

7. $\lim_{x \to a} c = c$

8. $\lim_{x \to a} x = a$

9. $\lim_{x \to a} x^n = a^n$ where n is a positive integer

10. $\lim_{x \to a} \sqrt[n]{x} = \sqrt[n]{a}$ where n is a positive integer
    (If n is even, we assume that $a > 0$.)

11. $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$ where n is a positive integer
    [If n is even, we assume that $\lim_{x \to a} f(x) > 0$.]

Example Evaluate the following limits.

(a) $\lim_{x\to 5}(2x^2-3x+4)$

(b) $\lim_{x\to -2}\dfrac{x^3+2x^2-1}{5-3x}$

Theorem [Direct Substitution Property]

If f is a polynomial or a rational function and a is in the domain of f, then

$$\lim_{x\to a} f(x) = f(a).$$

Recall Theorem If the functions f and g are equal for all x close to a, but not necessarily at $x=a$, then

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$$

whenever either limit exists.

Example Find $\lim_{x\to 1} g(x)$ where

$$g(x)=\begin{cases} x+1 & \text{if} \quad x\neq 1 \\ \pi & \text{if} \quad x=1 \end{cases}$$

Example Evaluate (a) $\lim_{h\to 0}\frac{(3+h)^2-9}{h}$ (b) $\lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2}$.

Theorem $\lim_{x\to a} f(x)=L$ if and only if $\lim_{x\to a^-} f(x)=L=\lim_{x\to a^+} f(x)$

Example Show that $\lim_{x\to 0}|x|=0$.

Example Show that $\lim_{x\to 0}\frac{|x|}{x}=0$ does not exist.

Example The greatest integer function is defined by $[\![x]\!] =$ the largest integer that is less than or equal to x.

Show that $\lim_{x \to 3} [\![x]\!]$ does not exist.

■ Other notations for $[\![x]\!]$ are $[x]$ and $\lfloor x \rfloor$. The greatest integer function is sometimes called the *floor function*.

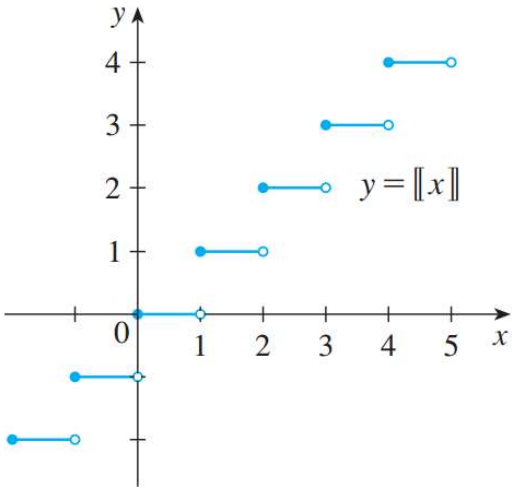


FIGURE 5
Greatest integer function

Theorem If $f(x) \le g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a, then

$$\lim_{x \to a} f(x) \le \lim_{x \to a} g(x)$$

Theorem [The Squeeze Theorem // the Sandwich Theorem]

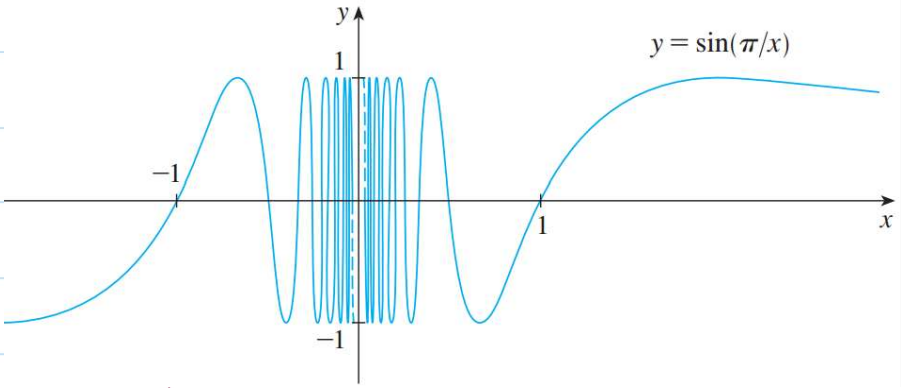
If $f(x) \le g(x) \le h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} g(x) = L$$

Example Evaluate $\lim_{x \to 0} x^2 \sin \frac{1}{x}$.



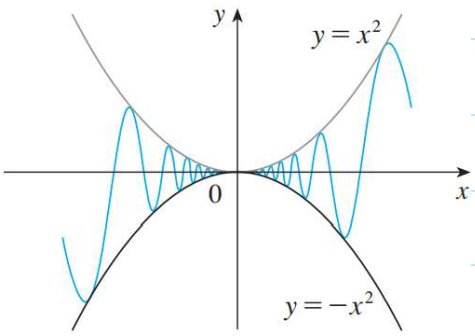


FIGURE 7
$y = x^2 \sin(1/x)$

§1.5 Continuity

Definition A function f is **continuous** at a number a if

$$\lim_{x \to a} f(x) = f(a)$$

Note A function is continuous at $x = a$ if

i. $f(a)$ is defined

ii. $\lim_{x \to a} f(x)$ exists

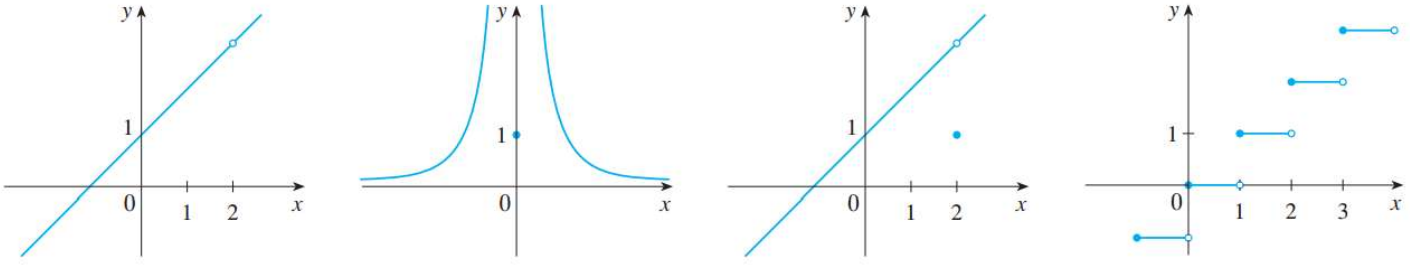
iii. $\lim_{x \to a} f(x) = f(a)$.

Definition A function is **discontinuous** at $x = a$ if it is not continuous at $x = a$.

[Types of Discontinuities]

Definition If $f(x)$ is discontinuous at a, then

1. f has a removable discontinuity at a
   if $\lim_{x \to a} f(x)$ exists as a real number, but it is different from $f(a)$.
2. f has a jump discontinuity at a
   if $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ both exist but $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$.
3. f has an infinite discontinuity at a
   if $\lim_{x \to a^-} f(x) = \pm\infty$ or $\lim_{x \to a^+} f(x) = \pm\infty$.



(a) $f(x) = \dfrac{x^2 - x - 2}{x - 2}$ (b) $f(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$ (c) $f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$ (d) $f(x) = [\![x]\!]$

FIGURE 3 Graphs of the functions in Example 2

2 DEFINITION A function f is **continuous from the right at a number** a if

$$\lim_{x \to a^+} f(x) = f(a)$$

and f is **continuous from the left at** a if

$$\lim_{x \to a^-} f(x) = f(a)$$

Example Let n be an any integer.
Determine whether the function $f(x) = [\![x]\!]$ is continuous from the left / the right at n.

Definition A function f is continuous on an interval if it is continuous at every number in the interval.
(If f is defined only on one side of an endpoint of the interval, we understand continuous at the endpoint to mean continuous from the right or continuous from the left.)

Example Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

4 THEOREM If f and g are continuous at a and c is a constant, then the following functions are also continuous at a:

1. $f + g$
2. $f - g$
3. cf
4. fg
5. $\frac{f}{g}$ if $g(a) \neq 0$

5 THEOREM

(a) Any polynomial is continuous everywhere; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$.

(b) Any rational function is continuous wherever it is defined; that is, it is continuous on its domain.

6 THEOREM The following types of functions are continuous at every number in their domains: polynomials, rational functions, root functions, trigonometric functions

Example Find $\lim_{x \to -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$.

Theorem [Limits of Composite Functions]

1. If f is continuous at b and $\lim_{x\to a} g(x) = b$, then

$$\lim_{x\to a} f(g(x)) = f\left(\lim_{x\to a} g(x)\right) = f(b).$$

2. If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ is given by $(f \circ g)(x) = f(g(x))$ is continuous at a.

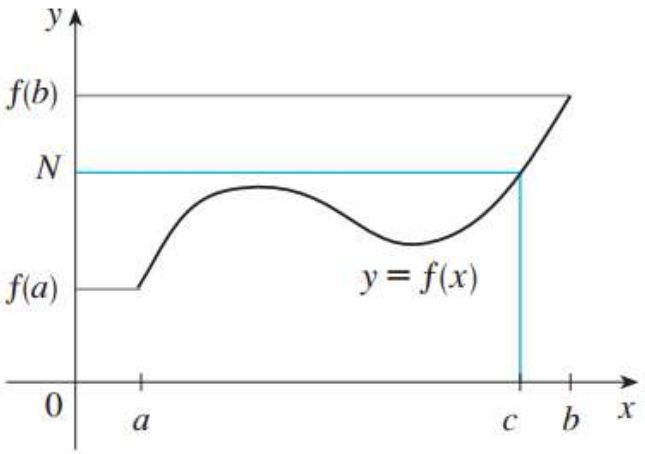
Example Where are the following functions continuous?

(a) $h(x) = \sin(x^2)$

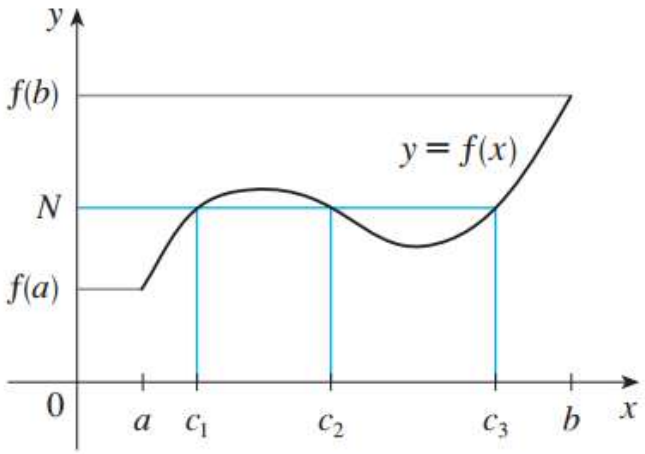
(b) $F(x) = \dfrac{1}{\sqrt{x^2+7}-4}$

Theorem [The Intermediate Value Theorem]

Suppose that f is continuous on the ***closed*** interval $[a,b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number $c \in (a,b)$ such that $f(c) = N$.



(a)



(b)

Example Show that there is a root(solution) of the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$

between 1 and 2.

§1.6 Limits Involving Infinity

[Infinite Limits]

1 DEFINITION The notation

$$\lim_{x \to a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a (on either side of a) but not equal to a.

2 DEFINITION The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a} f(x) = -\infty \qquad \lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

Example Find $\lim_{x \to 3^+} \frac{2x}{x-3}$ and $\lim_{x \to 3^-} \frac{2x}{x-3}$.

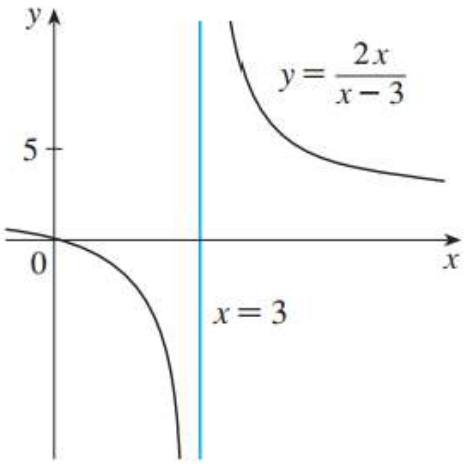


FIGURE 5

[Limits at Infinity]

3 DEFINITION Let f be a function defined on some interval (a, ∞). Then

$$\lim_{x \to \infty} f(x) = L$$

means that the values of $f(x)$ can be made as close to L as we like by taking x sufficiently large.

4 DEFINITION The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \to \infty} f(x) = L \qquad \text{or} \qquad \lim_{x \to -\infty} f(x) = L$$

Example Find $\lim_{x \to \infty} \frac{1}{x}$ and $\lim_{x \to -\infty} \frac{1}{x}$.

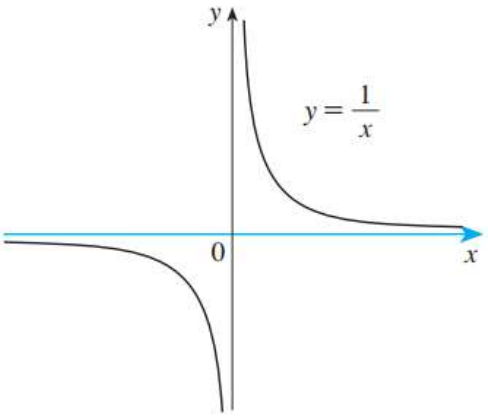


FIGURE 12

Theorem If n is a positive integer, then

$$\lim_{x \to \infty} \frac{1}{x^n} = 0 \quad \text{and} \quad \lim_{x \to -\infty} \frac{1}{x^n} = 0$$

Example Compute $\lim_{x\to\infty}(\sqrt{x^2+1}-x)$.

[Infinite Limits at Infinity]

$\lim_{x\to\infty} f(x)=\infty$: the values of $f(x)$ become large as x becomes large.

Example Find $\lim_{x\to\infty} x^3$ and $\lim_{x\to-\infty} x^3$.

Example Find $\lim_{x\to\infty}(x^2-x)$ and $\lim_{x\to\infty}\frac{x^2+x}{3-x}$.

Optional [Precise Definitions]

6 DEFINITION Let f be a function defined on some open interval that contains the number a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = \infty$$

means that for every positive number M there is a positive number δ such that

$$\text{if} \quad 0 < |x - a| < \delta \quad \text{then} \quad f(x) > M$$

7 DEFINITION Let f be a function defined on some interval (a, ∞). Then

$$\lim_{x \to \infty} f(x) = L$$

means that for every $\varepsilon > 0$ there is a corresponding number N such that

$$\text{if} \quad x > N \quad \text{then} \quad |f(x) - L| < \varepsilon$$

8 DEFINITION Let f be a function defined on some interval (a, ∞). Then

$$\lim_{x \to \infty} f(x) = \infty$$

means that for every positive number M there is a corresponding positive number N such that

$$\text{if} \quad x > N \quad \text{then} \quad f(x) > M$$